

Incident Report

Device ID:	DEV-001	Sector:	1
Device Name:	AEGIS-PUMP-01	Timestamp:	2026-05-09 06:49:17
Incident Signature:	cc60a8b3e28e198544a544c3e52f9e8a0b0c6a3bd42b48e6117ca1578171d993		
Trust Score:	14.9%	Attack Pattern:	Data Exfiltration (large payloads + high entropy)

Executive Summary

A critical anomaly was detected on the digital twin for the device. The system automatically generated this report to capture a snapshot of the observed behavior, analysis results, and recommended next steps for investigation and remediation.

Digital Twin Analysis

The LSTM autoencoder measured the reconstruction error and compared the incoming traffic features to the expected baseline. These discrepancies were used to compute a Trust Score and to highlight the most anomalous data attributes.

Metric	Value
Trust Score	14.9%
Reconstruction Error (MSE)	0.47143
Jensen-Shannon Divergence	0.40801

Top Anomaly Contributing Features

Feature	Delta
Entropy	0.700
Packet Size	0.600
Symmetry	0.600

Behavioral Timeline

- 12:19:15: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:18:53: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:18:30: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:17:55: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:17:37: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:17:29: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:17:24: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 12:17:18: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

12:17:11: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

12:17:06: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

Possible Attack Pattern

Data Exfiltration (large payloads + high entropy)

Risk Assessment

Based on the observed anomaly magnitude and the estimated severity, this incident is considered **CRITICAL**. The device should be treated as compromised until a full investigation confirms otherwise.

Recommended Remediation Actions

- Isolate the device from the network (quarantine).
- Collect full packet captures and system logs for post-incident analysis.
- Perform a firmware integrity check and verify configuration drift.
- Rotate credentials and revoke all active sessions for the device.
- Update IDS/IPS rules to detect similar behavior in the future.

Digital Evidence Snapshot

Baseline Features	Current Features
0.400, 0.500, 0.300, 0.600	1.000, 0.000, 1.000, 0.000

Incident Signature

cc60a8b3e28e198544a544c3e52f9eca0b8b6a3bd42b48e61b7c6f578171d993